



Africa Working Group – Meeting 2

HOST: Centre for Epidemic Response and Innovation (CERI)

DATE: March 04, 2025

TIME: 12:00 – 13:00 (SAST)

FACILIATORS: Dr Monika Moir and Dr Houriiyah Tegally

START TIME: 12:03 (SAST)

ATTENDEES: 35

PURPOSE

CLIMADE-Africa Scientific Planning for 2025: Uncovering Climate Change Impacts on Arbovirus Transmission Dynamics at the Intersection of Phylodynamics, Ecological Modelling and Machine Learning.

AGENDA

1. Welcome
2. Presentation and discussion led by Dr Houriiyah Tegally on the Consortium's scientific planning for 2025
3. Africa Stars Fellowship opportunities
4. Discussion and Feedback

DISCUSSION

- Dr Houriiyah Tegally presented a summary of the projects that will be conducted at CERI, focusing on the analytical and modeling aspects of climate-related research through a

CLIMADE funded project titled: "**Uncovering Climate Change Impacts on Arbovirus Transmission Dynamics at the Intersection of Phylodynamics, Ecological Modelling, and Machine Learning**". The presentation covered:

- An overview and summary of the project's aims that outline the analytical pipeline for understanding transmission dynamics in local and global contexts.
- Expected outcomes and deliverables.
- Project duration.
- Studies and current research outputs.
- Benefit of this research and potential for public health responses.
- Gaps in data and research were highlighted, such as vector-related research, including the presence, occurrences, abundance and vector-viral competency studies.
- Introduction to the new methodologies that will be used and data types, such as the use of migration data, machine learning and AI for modelling disease transmission and epidemics.
- Potential collaborations were encouraged.
- New repository for data analysis (currently private but will be made publicly available).
- The Mastercard Africa Stars Fellowship programme
 - Internship opportunities at CERI and Institute Pasteur de Dakar.
 - Fellowships are available for 4 months and 12 months, covering various scientific areas.
 - Applications end on the 31st of March 2025.

Discussion points and Call for Collaboration

- Animals and Non-Human Primates (NHP) that have been implicated in most of the outbreaks. In Africa, we do not have many studies that focus on animals—currently there is no specific project that will look at animal/NHP/ reservoir hosts, however, this is an important aspect to consider since in West Africa many outbreaks are often related to spillover events. Reservoir host distribution can also be included in modelling disease transmission.
- What kind of information is most needed? – Studies on the viral presence in the mosquito pools/vector pools and survey studies are needed to identify vector presence as there are

studies that look at environmental suitability for vectors without presence and abundance data.

- Is there funding support for data collection (field data)? –There is no specific funding in the discussed project for this work. We can work together to find grants that will help fund these initiatives and studies.
- Are there opportunities for collaboration? Yes, if you are interested in collaboration, please contact the CERI-CLIMADE team.

ADJOURNMENT AND CLOSING POINTS

Dr Houriiyah Tegally encouraged applications for the Africa Stars Fellowship programme and collaboration within the CLIMADE consortium. The meeting was adjourned at 13:13 (SAST).

NEXT MEETING

Presentation by Dr. Etienne Frumence from the University Hospital of Reunion Island on Molecular Epidemiology of Arboviruses in Reunion Island (2012–2025): Insights from Genomic Data.

Date: March 25, 2025, at 12:00 (SAST)

Platform: Online (MS Teams)

Submitted by: Yajna Ramphal

Approved by: Monika Moir